EXP – 5 : SCHEDULING ALGORITHMS (PRIORITY SCHEDULING)

```
[rene@localhost ~]$ ls
abc.sh     Desktop    filename.txt  newfile.c   producerconsumer  text
a.out      Documents  file.sh       page        Public            text.c
bank       Downloads  firstfit.c    p.c         rec               Videos
bankers    e.sh       first.sh      Pictures    scheduling        xyz
bankers.c  exec.c     memory        pid.c       sjf.c
cd         f.c        Music         pro         system_calls
ch         fcfs.c     name.sh       producer.c  Templates
[rene@localhost ~]$ cd scheduling
[rene@localhost scheduling]$ vi priority.c
[rene@localhost scheduling]$
```

```c
#include<stdio.h>
int main(){
        int n,completed =0,time=0;
        printf("Enter number of processes: \n");
        scanf("%d",&n);
        int at[n], bt[n],priority[n],done[n];
        int ct[n],tat[n],wt[n];
        float avgtat = 0,avgwt = 0;
        for(int i=0;i<n;i++)
        {
                printf("Enter AT,BT and Priority for P%d: \n", i + 1);
                scanf("%d %d %d",&at[i],&bt[i],&priority[i]);
                done[i] = 0;
        }
        while(completed < n)
        {
                int idx = -1;
                int highPriority = 9999;
                        for(int i=0;i<n;i++)
                        {
                                if(at[i] <= time && done[i] ==0)
                                {
                                        if(priority[i] < highPriority)
                                        {
                                                highPriority = priority[i];
                                                idx = i;
                                        }
                                }
                        }
                if(idx != -1)
                {
                         time += bt[idx];
                         ct[idx] = time;
                         tat[idx] = ct[idx] - at[idx];
                         wt[idx] = tat[idx] - bt[idx];
                         done[idx] = 1;
                         completed++;
                }
                else
                {
                        time++;
                }
        }
        for(int i=0;i<n;i++)
        {
                                                                        1,17          Top
```

```c
        {
                printf("Enter AT,BT and Priority for P%d: \n", i + 1);
                scanf("%d %d %d",&at[i],&bt[i],&priority[i]);
                done[i] = 0;
        }
        while(completed < n)
        {
                int idx = -1;
                int highPriority = 9999;
                        for(int i=0;i<n;i++)
                        {
                                if(at[i] <= time && done[i] ==0)
                                {
                                        if(priority[i] < highPriority)
                                        {
                                                highPriority = priority[i];
                                                idx = i;
                                        }
                                }
                        }
                if(idx != -1)
                {
                         time += bt[idx];
                         ct[idx] = time;
                         tat[idx] = ct[idx] - at[idx];
                         wt[idx] = tat[idx] - bt[idx];
                         done[idx] = 1;
                         completed++;
                }
                else
                {
                        time++;
                }
        }
        for(int i=0;i<n;i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }
        printf("\nAverage Turnaround Time: %.2f\n",avgtat /n);
        printf("Average Waiting Time: %.2f\n",avgwt / n);
        return 0;
}
                                                                                                                                52,1          Bot
```

```
[rene@localhost ~]$ ls
abc.sh     Desktop    filename.txt  newfile.c   producerconsumer  text
a.out      Documents  file.sh       page        Public            text.c
bank       Downloads  firstfit.c    p.c         rec               Videos
bankers    e.sh       first.sh      Pictures    scheduling        xyz
bankers.c  exec.c     memory        pid.c       sjf.c
cd         f.c        Music         pro         system_calls
ch         fcfs.c     name.sh       producer.c  Templates
[rene@localhost ~]$ cd scheduling
[rene@localhost scheduling]$ vi priority.c
[rene@localhost scheduling]$ gcc priority.c
[rene@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter AT,BT and Priority for P1:
0 5 10
Enter AT,BT and Priority for P2:
1 4 20
Enter AT,BT and Priority for P3:
2 2 30
Enter AT,BT and Priority for P4:
4 1 40

Average Turnaround Time: 7.50
Average Waiting Time: 4.50
[rene@localhost scheduling]$
```